

Pod Name	Node	GPU	Startup (sec)
qwen-qwe-c9c8568d-en3-0-6b-decode-7bbc774556-l2dgv	pokprod-b93r43s0	NVIDIA-H100-80GB-HBM3	172.0s
qwen-qwe-c9c8568d-en3-0-6b-decode-7bbc774556-gccwx	pokprod-b93r43s2	NVIDIA-H100-80GB-HBM3	180.0s
qwen-qwe-c9c8568d-en3-0-6b-decode-7bbc774556-vdh17	pokprod-b93r44s3	NVIDIA-H100-80GB-HBM3	N/A (Missing Ready Data)

EPP Configuration (Feature Gates & Scorer Weights)

apiVersion: inference.networking.x-k8s.io/v1alpha1
kind: EndpointPickerConfig
featureGates:
- flowControl
plugins:
- type: queue-scorer
- type: kv-cache-utilization-scorer
- type: prefix-cache-scorer
schedulingProfiles:
- name: default
 plugins:
 - pluginRef: queue-scorer
 weight: 2
 - pluginRef: kv-cache-utilization-scorer
 weight: 2
 - pluginRef: prefix-cache-scorer
 weight: 3

Benchmark Load Generator Configuration

Profile: chatbot_synthetic.yaml
target: http://infra-llmdbench-inference-gateway-istio.wva-bench-test.svc.cluster.local:80
model: Qwen/Qwen3-0.6B
request_type: text_completions
profile: constant
rate: [1,2,4,8]
max_seconds: 600
random_seed: 42
data:
 prompt_tokens_min: 10
 prompt_tokens_max: 8192
 prompt_tokens: 4096
 prompt_tokens_stdev: 2048
 output_tokens_min: 10
 output_tokens_max: 2048
 output_tokens: 1024
 output_tokens_stdev: 512
 samples: 1000

Autoscaling Configuration (HPA & VA)

Variant (VA) [workload-variant-autoscaler-va]: Min Replicas = 1, Max Replicas = 10, Cost Factor = 10.0
HorizontalPodAutoscaler (HPA) [workload-variant-autoscaler-hpa]: Min Replicas = 1, Max Replicas = 10
 > ScaleUp: Stabilization Window = 240s | Policies = [{"periodSeconds": 150, "type": "Pods", "value": 10}]
 > ScaleDown: Stabilization Window = 240s | Policies = [{"periodSeconds": 150, "type": "Pods", "value": 10}]

True Serving Capacity Analysis (GuideLLM results.json)

Rate: 1.0 RPS	Achieved: 0.99 RPS	Errors: 0.00 RPS	Tokens/s: 5143.67
Rate: 2.0 RPS	Achieved: 1.97 RPS	Errors: 0.00 RPS	Tokens/s: 10117.24
Rate: 4.0 RPS	Achieved: 3.87 RPS	Errors: 0.07 RPS	Tokens/s: 20049.52
Rate: 8.0 RPS	Achieved: 5.23 RPS	Errors: 1.11 RPS	Tokens/s: 26919.35

=> Estimated True Serving Capacity: ~1.97 RPS (Highest successful rate with 0 errors)

Understanding GuideLLM Metrics:

- Target Rate (RPS): The configured constant request rate the load generator attempts to send.
- Achieved RPS: The actual rate of requests that completed successfully with a full response.
- Error RPS: The rate of requests that failed (e.g., HTTP 5xx) or were strictly dropped/incomplete.
- True Serving Capacity: Evaluated as the highest Achieved RPS recorded just before the system reaches hardware saturation and begins generating Error RPS (dropped requests).

Autoscaling Run Score (Lower is Better)

Worst-Case P99 TTFT: 78179.15 ms
Worst-Case P99 ITL: 31.99 ms
Average Replicas: 1.21

Target SLAs: TTFT = 50.0ms | ITL = 50.0ms

Formula Engine:

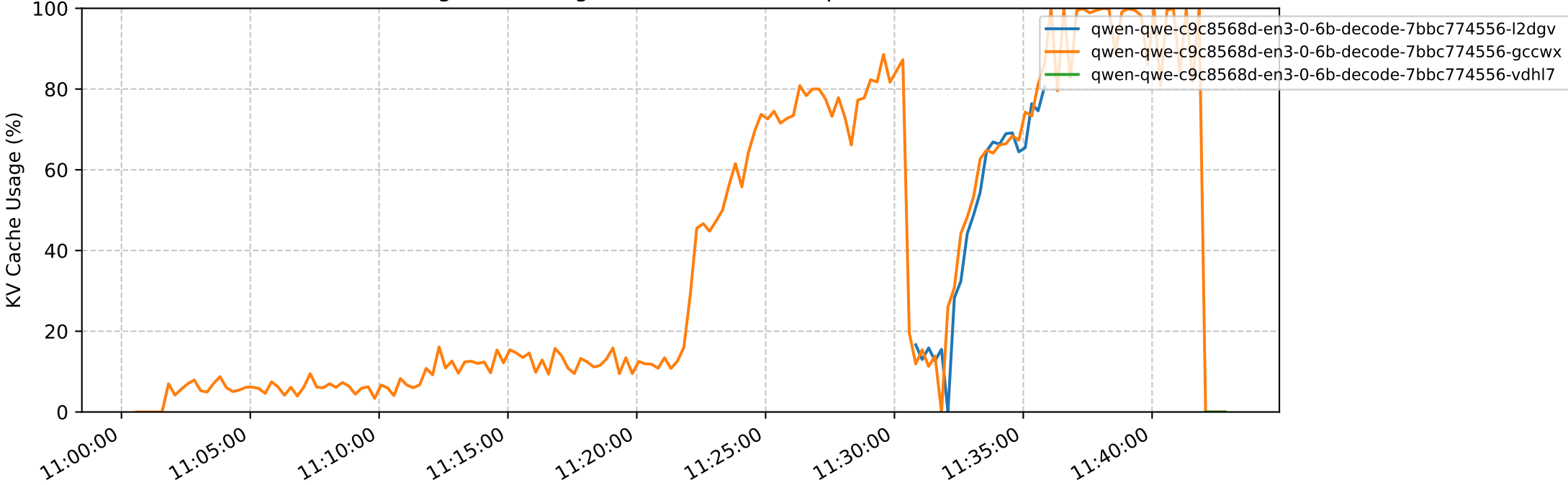
1. TTFT Penalty = (Actual P99 TTFT) / (SLA TTFT)
 2. ITL Penalty = (Actual P99 ITL) / (SLA ITL)
 3. Cost Factor = VA Custom Resource declared Cost (10.0)
 4. Avg Replicas Factor = Average running decode pods (1.21)
- Autoscaling Score = Avg_Replicas × Cost_Factor × (TTFT_Penalty + ITL_Penalty)

Latency Penalty Subtotal = (78179.15/50.0) + (31.99/50.0) = 1564.22
Resource Multiplier = 1.21 × 10.0 = 12.12
=> Final Score = 12.12 × 1564.22 = 18954.70

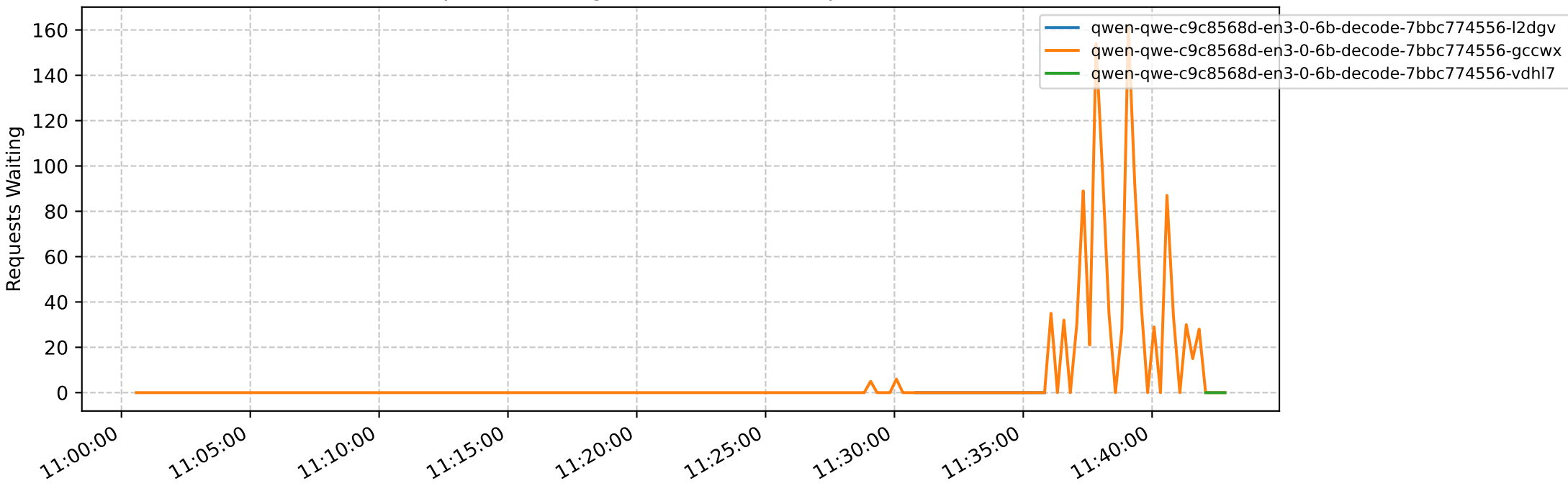
GuideLLM Benchmark Inter-Phase Delay Analysis

Phase 0 (benchmark_report_results.json_0.yaml): Start = 1773770494.58, Stop = 1773771094.58 (Duration: 600.0s)
Gap between Phase 0 and 1: 3.67 seconds
Phase 1 (benchmark_report_results.json_1.yaml): Start = 1773771098.25, Stop = 1773771698.25 (Duration: 600.0s)
Gap between Phase 1 and 2: 5.56 seconds
Phase 2 (benchmark_report_results.json_2.yaml): Start = 1773771703.81, Stop = 1773772303.81 (Duration: 600.0s)
Gap between Phase 2 and 3: 9.33 seconds
Phase 3 (benchmark_report_results.json_3.yaml): Start = 1773772313.14, Stop = 1773772913.14 (Duration: 600.0s)

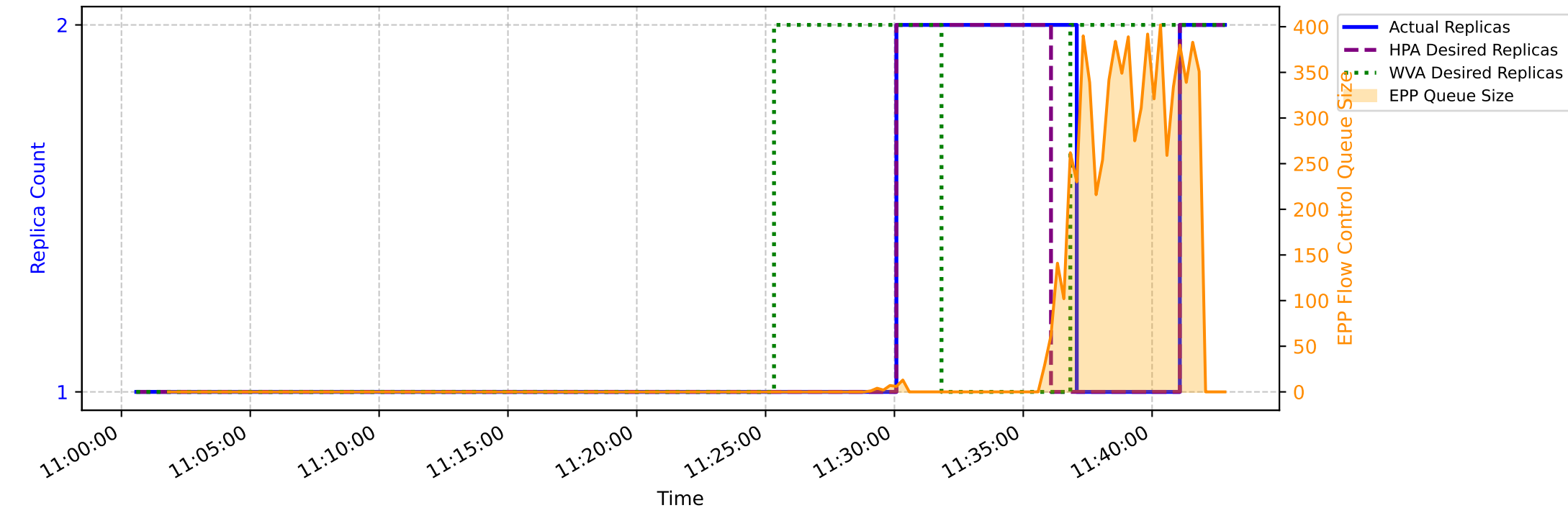
KV Cache Usage Percentage Over Time (Namespace: wva-bench-test)



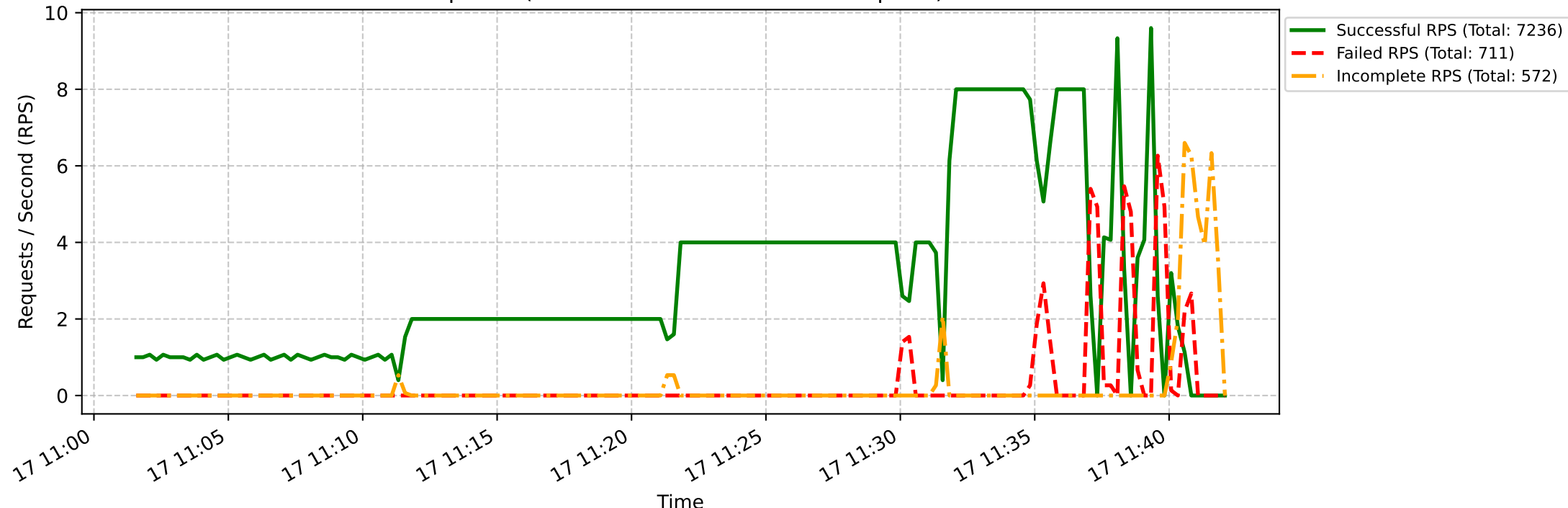
Number of Requests Waiting Over Time (Namespace: wva-bench-test)



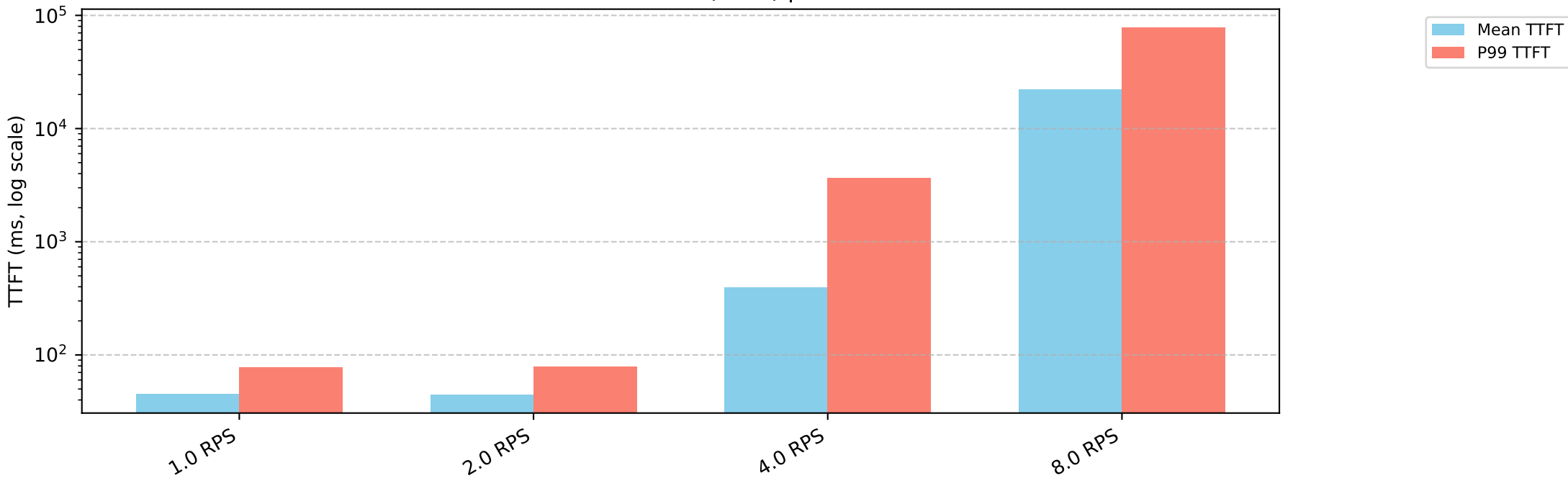
Decode Replica Count & EPP Queue Over Time (Namespace: wva-bench-test)



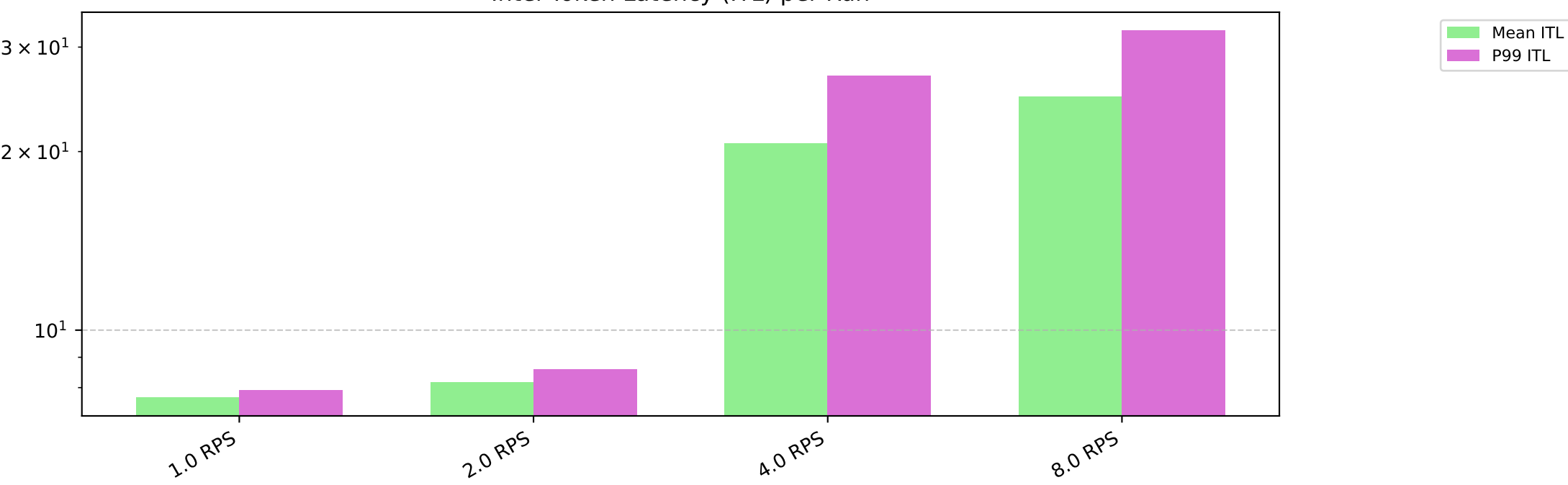
GuideLLM Requests (Succeeded vs Failed vs Incomplete) Over Time



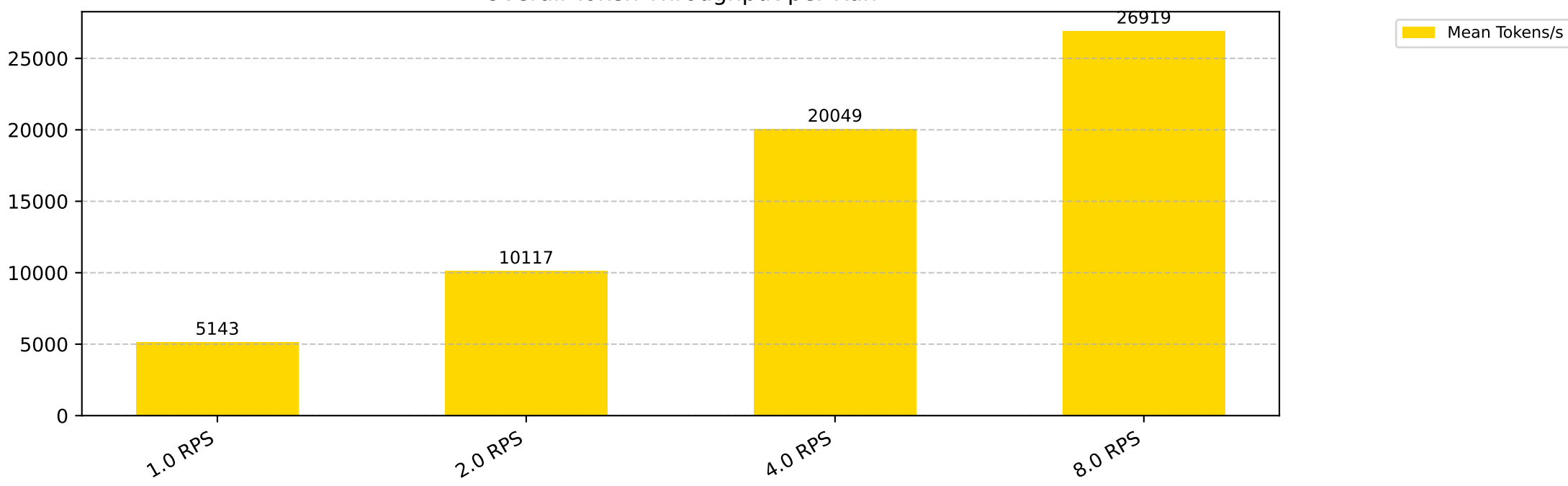
Time To First Token (TTFT) per Run



Inter-Token Latency (ITL) per Run



Overall Token Throughput per Run



Request Concurrency vs End-to-End Latency Profile

